

# Solutions to "Algebraic Geometry and Arithmetic Curves"

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## 1 Some topics in commutative algebra

### 1.1 Tensor products

**Exercise 1.1.1.** Let  $\{M_i\}_{i \in I}, \{N_j\}_{j \in J}$  be two families of modules over a ring  $A$ . Then

$$(\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j).$$

*Proof.* There is a chain of natural isomorphisms:

$$\begin{aligned} & (\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \\ & \cong \oplus_{i \in I} (M_i \otimes_A (\oplus_{j \in J} N_j)) \\ & \cong \oplus_{j \in J} \oplus_{i \in I} (M_i \otimes_A N_j) \\ & \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j). \end{aligned}$$

□

**Exercise 1.1.2.** There is a unique  $A$ -linear map

$$f : \text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that  $f(u \otimes v) = u \otimes v$ .

*Proof.* Consider the map  $g : \text{Hom}_A(M, M') \times \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$  sending  $(u, v)$  to  $u \otimes v$ . It is clearly bilinear, so factors uniquely through  $\text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N')$ . □

**Exercise 1.1.3.** Let  $M, N$  be  $A$ -modules and let  $i : M' \rightarrow M, j : N' \rightarrow N$  be submodules of  $M$  and  $N$ , respectively. Then there is a canonical isomorphism

$$(M/M') \otimes_A (N/N') \cong (M \otimes_A N) / (\text{Im } i_N + \text{Im } j_N).$$

In particular,  $(\mathbf{Z}/n\mathbf{Z}) \otimes_{\mathbf{Z}} (\mathbf{Z}/m\mathbf{Z}) = \mathbf{Z}/l\mathbf{Z}$ , where  $l = \gcd(m, n)$ .

*Proof.* Observe that

$$(M \otimes_A N)/(\text{Im } i_N + \text{Im } j_N) \cong ((M \otimes_A N)/(\text{Im } i_N))/(\text{Im } j_{M/M'}),$$

so Corollary 1.13 gives us a chain of isomorphisms

$$\begin{aligned} & ((M \otimes_A N)/(\text{Im } i_N))/(\text{Im } j_{M/M'}) \\ & \cong ((M/M') \otimes_A N)/(\text{Im } j_{M/M'}) \\ & \cong (M/M') \otimes_A (N/N'). \end{aligned}$$

For the last statement, just recall that  $(m) + (n) = (l)$ . □

**Exercise 1.1.4.** Let  $M, N$  be  $A$ -modules and let  $B, C$  be  $A$ -algebras.

- (a) If  $M$  and  $N$  are finitely generated over  $A$ , then so is  $M \otimes_A N$ .
- (b) If  $B$  and  $C$  are finitely generated over  $A$ , then so is  $B \otimes_A C$ .
- (c) The converses of (a) and (b) are in general false.

*Proof.* Parts (a) and (b) are obvious. For part (c), just notice that  $\mathbf{Q} \otimes_{\mathbf{Z}} \mathbf{Z}/2\mathbf{Z}$  is the zero ring, but  $\mathbf{Q}$  is not finitely generated over  $\mathbf{Z}$ . □

**Exercise 1.1.5.** Let  $\rho : A \rightarrow B$  be a ring homomorphism,  $M$  an  $A$ -module, and  $N$  a  $B$ -module. There is a canonical isomorphism of  $A$ -modules

$$\text{Hom}_A(M, \rho_* N) \cong \text{Hom}_B(\rho^* M, N).$$

*Proof.* Let  $f : \text{Hom}_A(M, \rho_* N) \rightarrow \text{Hom}_B(\rho^* M, N)$  be the map sending  $u$  to the map  $f(u)$  s.t.  $f(u)(m \otimes b) = bu(m)$ . This is well-defined because if  $n \otimes c = m \otimes b$ , we may assume that there is some  $a \in A$  s.t.  $an = m$  and  $\rho(a)b = c$  and so  $f(u)(n \otimes c) = cu(n) = \rho(a)bu(n) = bu(m)$ .

In addition, define  $g : \text{Hom}_B(\rho^* M, N) \rightarrow \text{Hom}_A(M, \rho_* N)$  to be the map s.t.  $g(u)(m) = u(m \otimes 1)$ . This is immediately seen to be well-defined, and we contest that this is an inverse for  $f$ . Indeed,  $f(g(u))(m \otimes b) = bg(u)(m) = bu(m \otimes 1) = u(m \otimes b)$  and  $g(f(u))(m) = f(u)(m \otimes 1) = u(m)$ . □

**Exercise 1.1.6.** Let  $(N_i)_{i \in I}$  be a direct system of modules. Then for any  $A$ -module  $M$ , there exists a canonical isomorphism  $\varinjlim (N_i \otimes_A M) \cong (\varinjlim N_i) \otimes_A M$ .

*Proof.* Tensor-hom.

Just kidding, it follows from the fact that  $\varinjlim (N_i \times M)$  is canonically isomorphic to  $(\varinjlim N_i) \times M$ , and that a bilinear map over the latter must then induce a cocone of bilinear maps. □

**Exercise 1.1.7.** Let  $B$  be an  $A$ -algebra, and let  $M, N$  be  $B$ -modules. Then there exists a canonical surjection  $M \otimes_A N \rightarrow M \otimes_B N$ .

*Proof.* Do I really need to spell it out? □

## 1.2 Flatness

**Exercise 1.2.1.** Let  $M$  be an  $A$ -module and let  $I \subseteq \text{Ann}(m)$  be an ideal.

- (a) There is a natural  $A/I$ -module structure on  $M$ , and  $M \cong M \otimes_A A/I$ .
- (b) Let  $N$  be another  $A$ -module such that  $I \subseteq \text{Ann}(N)$ . The canonical homomorphism  $M \otimes_A N \rightarrow M \otimes_{A/I} N$  is an isomorphism.

*Proof.* For (a), observe that if  $x \in I$ , then  $am = (a + x)m$  for any  $a \in A$  and  $m \in M$ . This implies that the multiplication of elements of  $M$  by cosets  $a + I$  is well defined, hence we have an  $A/I$ -module structure on  $M$ . To see that this is the extension of scalars, recall that  $M \otimes_A A/I \cong M/(IM)$ , which is just  $M$ .

Part (b) follows from a chain of isomorphisms

$$\begin{aligned} M \otimes_A N &\cong (M \otimes_A A/I) \otimes_{A/I} N \\ &\cong M \otimes_{A/I} N, \end{aligned}$$

where the first isomorphism comes from Corollary 1.11 and the second comes from part (a).  $\square$

**Exercise 1.2.2.** Let  $\rho : A \rightarrow B$  be a ring homomorphism,  $S$  a multiplicative subset of  $A$ , and  $T = \rho(S)$ . Then  $T$  is a multiplicative subset of  $B$ , and  $T^{-1}B \cong B \otimes_A S^{-1}A$ .

*Proof.* The first part is obvious. To see the last statement, let  $\iota : B \rightarrow T^{-1}B$  be the canonical embedding. Since  $\iota$  sends elements of  $T = \rho(S)$  to units, there is a canonical homomorphism  $\phi : S^{-1}A \rightarrow T^{-1}B$ . Since  $\otimes_A$  is a pushout in the category of  $A$ -algebras, there is a unique homomorphism  $B \otimes_A S^{-1}A \rightarrow T^{-1}B$  such that this diagram commutes:

$$\begin{array}{ccc} S^{-1}A \otimes B & \longleftarrow & S^{-1}A \\ \uparrow & \searrow & \downarrow \\ B & \longrightarrow & T^{-1}B. \end{array}$$

This is an isomorphism because of universal property of localizations.  $\square$

**Exercise 1.2.3.** Nakayama's lemma does not hold for modules that are not finitely generated.

*Proof.* Consider  $\mathbf{Q}$  as a module over  $\mathbf{Z}_{(p)}$ .  $\square$

## 1.3 Formal completion

# 2 General properties of schemes

## 2.1 Spectrum of a ring

**Exercise 2.1.1.** For  $k$  a field,  $\text{Spec}(k[[T]])$  is the space  $\{\xi, 0\}$  with generic point  $\xi$  and closed point 0.

*Proof.* It is easy to see that any power series with a non-zero constant term is a unit. Furthermore, if we have  $f = \sum a_n T^n$  with  $a_N = 1$  for some  $N$  and  $a_n = 0$  whenever  $n \leq N$ , then we can pull out a common factor of  $T^N$ . We see that  $f$  is then a unit multiple of  $T^N$  and  $(f) = (T^N)$ . Clearly, this is only prime when  $N = 1$ . It is easy to see that  $k[[T]]$  is an entire ring, so the proposition is immediate.  $\square$

**Exercise 2.1.2.** Let  $\varphi : A \rightarrow B$  be a homomorphism of finitely generated algebras over a field  $k$ . The image of a closed point under  $\text{Spec } \varphi$  is a closed point.

*Proof.* If  $\text{Spec } \varphi$  takes the maximal ideal  $\mathfrak{m}$  into the ideal  $\mathfrak{p}$ , we get a diagram

$$k \rightarrow A/\mathfrak{p} \xrightarrow{\bar{\varphi}} B/\mathfrak{m}.$$

By Corollary 1.12,  $B/\mathfrak{m}$  is a finite algebraic extension of  $k$ , so  $B/\mathfrak{m}$  (resp.  $A/\mathfrak{p}$ ) is finite over  $A/\mathfrak{p}$  (resp.  $k$ ). It then suffices to prove that  $A/\mathfrak{p}$  is a field. Any  $a \in A$  has an associated irreducible polynomial  $P = \sum a_i X^i \in k[x]$  such that  $P(a) = 0$ . This polynomial must have a nonzero constant term, so we have

$$-\frac{1}{a_0} \sum_{i>0} a_i a^i = 1,$$

but the polynomial on the left has no constant term and we can pull out a factor of  $a$ .  $\square$

**Exercise 2.1.3.** Let  $k = \mathbf{R}$  be the field of real numbers and let  $A = k[X, Y]/(X^2 + Y^2 + 1)$ . Let  $x, y$  be the respective images of  $X, Y$  in  $A$ .

- (a) Let  $\mathfrak{m}$  be a maximal ideal of  $A$ . Then there are  $a, b, c, d \in k$  such that  $x^2 + ax + b, y^2 + cy + d \in \mathfrak{m}$ . Furthermore,  $\mathfrak{m}$  is generated by a single element  $f = \alpha x + \beta y + \gamma$  with at least one of  $\alpha, \beta$  nonzero.
- (b) The map  $(\alpha, \beta, \gamma) \mapsto (\alpha x + \beta y + \gamma)A$  establishes a bijection between the subset  $\mathbf{P}(k^3) \setminus \{(0, 0, 1)\}$  of the projective space  $\mathbf{P}(k^3)$  and the set of maximal ideals of  $A$ .
- (c) Let  $\mathfrak{p}$  be a non-maximal prime ideal of  $A$ . Then the canonical homomorphism  $\varphi : k[X] \rightarrow A$  is finite and injective, and  $k[X] \cap \mathfrak{p} = 0$ . Let  $g \in \mathfrak{p}$  and let  $g^n + a_{n-1}g^{n-1} + \dots + a_0 = 0$  be an integral equation with  $a_i \in k[X]$ . Then  $a_0 = 0$ , and so  $\mathfrak{p} = 0$ .

*Proof.* First, notice that  $A/\mathfrak{m}$  is isomorphic to  $\mathbf{C}$ , since  $X^2 + Y^2 + 1$  does not split in  $k$ . Then the existence of  $a, b, c, d$  follows from the fact that any complex number satisfies a monic quadratic equation. Since  $\mathbf{C}$  is degree 2 over  $k$ , there are  $\alpha, \beta, \gamma$ , not all zero, such that

$$\alpha x' + \beta y' + \gamma = 0,$$

where  $x'$  and  $y'$  are the respective images of  $x$  and  $y$ . Since not all of them are zero, we may assume that  $\alpha$  is nonzero. Then we get an equation for  $x'$  in terms of  $y'$ , which we can then substitute into the equation  $x'^2 + y'^2 + 1 = 0$ . This new equation

cannot split in  $k$  because that would imply a real root of  $x'^2 + y'^2 + 1 = 0$ , so we have that  $\mathfrak{m} = fA$ , where  $f$  is the ideal corresponding to the linear equation in  $x'$  and  $y'$ .

Part (b) holds, since a solution to  $X^2 + Y^2 + 1 = 0$  is determined (up to conjugacy) by an arbitrary complex number, which itself determines a linear equation with real coefficients.

That  $\varphi$  is finite and injective is clear, so consider the induced homomorphism  $\bar{\varphi} : k[X]/\varphi^{-1}(\mathfrak{p}) \rightarrow A/\mathfrak{p}$ . Notice that  $B = k[X]/\varphi^{-1}(\mathfrak{p})$  is not a field, since  $\bar{\varphi}$  is still finite and injective, which implies that  $A/\mathfrak{p}$  is a field as well. Then  $B = k[X]$  and  $\mathfrak{p} \cap k[X] = 0$ . Finally, in the integral equation for  $g$ , notice that  $a_0$  must be a multiple of  $g$ , hence 0. Then the equation has a common factor of  $g$  which we may cancel, which gives a new equation with constant term  $a_1$ . This repeats indefinitely and so  $g = 0$ .  $\square$

**Exercise 2.1.4.** Let  $A$  be a ring.

- (a) Let  $\mathfrak{p}$  be a minimal prime ideal of  $A$ . Then every element of  $\mathfrak{p}A_{\mathfrak{p}}$  is nilpotent. In particular, every element of  $\mathfrak{p}$  is a zero divisor.
- (b) If  $A$  is reduced, then any zero divisor in  $A$  is an element of a minimal prime ideal. This is not true in general if  $A$  is not reduced.

*Proof.* Part (a) is obvious.

For (b), let  $x$  be any zero divisor and let  $\mathfrak{p}$  be a prime ideal minimal among those that contain  $x$ . If  $\mathfrak{p}$  is minimal, we are done. Otherwise, suppose that  $\mathfrak{q} \subset \mathfrak{p}$  is a minimal prime ideal not containing  $x$ . Since  $A$  is reduced, we must have  $\sqrt{(x)} \cap \text{Ann}(x) = \{0\}$ , so  $\mathfrak{q}$  must contain a nonzero element  $x' \in \text{Ann}(x)$ . Localizing by  $\mathfrak{q}$ , we see that  $x$  is sent to 0 and  $x'$  is sent to a nilpotent. This implies that there is some positive integer  $n$  and some element  $s \in A \setminus \mathfrak{q}$  such that  $s(x'^n - x) = 0$ , and it is easy to see that this means that  $sx^2 = 0 \in \mathfrak{q}$ , so  $x^2 \in \mathfrak{q}$ , contradiction.  $\square$

The fact that this is false for  $A$  not reduced comes is immediate from the fact that there can be non-nilpotent zero divisors in a ring.

**Exercise 2.1.5.** Let  $k$  be a field. Let  $\mathfrak{m}$  be a maximal ideal of  $k[T_1, \dots, T_n]$ .

- (a) Let  $P_1(T_1)$  be a generator of  $\mathfrak{m} \cap k[T_1]$  and  $k_1 = k[T_1]/(P_1)$ . Then we have an exact sequence

$$0 \rightarrow P_1 k[T_1, \dots, T_n] \rightarrow k[T_1, \dots, T_n] \rightarrow k_1[T_2, \dots, T_n] \rightarrow 0.$$

- (b) There exist  $n$  polynomials  $P_1(T_1), P_2(T_1, T_2), \dots, P_n(T_1, \dots, T_n)$  such that  $k[T_1, \dots, T_i] \cap \mathfrak{m} = (P_1, \dots, P_i)$  for all  $i \leq n$ . In particular,  $\mathfrak{m}$  is generated by  $n$  elements.

*Proof.* Part (a) is obvious. In part (b), we proceed by induction. The statement is obvious in the single-variable case, so suppose that it holds for  $n - 1$ , with  $n$  an integer at least 2. Let  $k_1$  be as in (a) and observe that it is a field, so the hypotheses and the exact sequence from (a) give a generating set  $(P_2, \dots, P_n)$  for the image of  $\mathfrak{m}$  in  $k_1[T_2, \dots, T_n]$ . This pulls back to a set of generators  $(P_1, \dots, P_n)$  for  $\mathfrak{m}$ , and it is easy to see that the desired properties hold.  $\square$

**Exercise 2.1.6.** Any closed subspace of a quasicompact space is itself quasicompact. If  $A$  is a ring,  $\text{Spec}(A)$  is quasicompact.

*Proof.* Let  $X$  be a topological space and let  $Y \subseteq X$  be any closed subset. If  $\{U_i\}$  is an open covering of  $Y$ ,  $\{U_i \cup \complement_X Y\}$  is an open covering of  $X$ . Taking a finite subcovering of this extended covering yields a finite subcovering of the original covering.

For the second statement, suppose that  $\{U_i\}$  is an open covering of  $X = \text{Spec } A$ . For each  $i$ , let  $J_i$  be the ideal such that  $\complement_X U_i = V(J_i)$ . We see that  $\emptyset = \bigcap V(J_i) = V(\sum J_i)$ , so there is a finite family of  $f_k \in J_k$  such that  $\sum f_k = 1$ . Then  $\sum J_k = 1$  and  $\{U_k\}$  is a finite subcovering.  $\square$

The next exercise is a bit open ended in character, so the proof label is changed to reflect that.

**Exercise 2.1.7.** Describe the closed subsets of  $\text{Spec}(\mathbb{Z}[T])$ . Show that they are not all principal.

*Answer.* The closed subsets of  $\text{Spec}(\mathbb{Z}[T])$  can be broken up into three families: points, lines, and the generic point. The points are simply maximal ideals; in  $\mathbb{Z}[T]$  these are ideals of the form  $(p, P)$ , where  $p$  is a non-zero prime number and  $P$  is an irreducible polynomial. The lines are just principal ideals; I call them lines because they determine a family of points that looks kind of like a line. Finally, there is the generic point  $(0)$ . The fact that the closed subsets are not all principal follows from the fact that the maximal ideals have 2 generators.  $\square$

## 2.2 Ringed topological spaces

Let  $X$  be a topological space.

**Exercise 2.2.1.** Let  $A$  be a non-trivial abelian group and let  $\mathcal{A}_X$  be the constant presheaf of  $A$  on  $X$ . The associated sheaf  $\mathcal{A}_X^1$  is the sheaf of locally constant functions valued in  $A$ . In particular,  $\mathcal{A}_X$  is a sheaf if and only if every non-empty open subset of  $X$  is connected.

*Proof.* Let  $\mathcal{F}$  be the sheaf of locally constant  $A$ -valued functions on  $X$  and let  $\varphi : \mathcal{A}_X \rightarrow \mathcal{F}$  be the evident morphism. It is easy to see that the stalks of  $\mathcal{F}$  are all  $A$ . Let  $\mathcal{G}$  be any sheaf and let  $\alpha : \mathcal{A}_X \rightarrow \mathcal{G}$  be any morphism. If  $f$  is a locally constant function on  $U$ , we may split  $U$  into disjoint open sets  $U_a = f^{-1}(a)$  for  $a \in f(U)$ . We define  $\tilde{\alpha}(U) : \mathcal{F} \rightarrow \mathcal{G}$  to be the map sending  $f$  to the regluing of

the sections  $\alpha(U)(a)|_{U_a}$ . It is easy to see that this defines a morphism of sheaves and that  $\tilde{\alpha} \circ \varphi = \alpha$ . Since  $\varphi$  is surjective,  $\tilde{\alpha}$  is the unique such morphism.  $\square$